Split-prime pattern and the ECDLP: a scalar-only obstruction, an independent recount, and the routes that stay open

research/split_prime_ecdlp_20261107 · 2026-11-07 · branch `claude/novel-ecdlp-algorithm-iams4j`

Status: derived obstruction plus a native recount; no algorithm claim. **Result class:** boundary study; lead closure for the scalar / generic route only, not a cost reduction for any ECDLP method. `S` and end-to-end cost are unset; no leaderboard, benchmark or rho comparison is touched. Claim record for the recount: `PCC-20261107-3403f9` (kind statement; standing unverified until CI signs it) [pcc:unverified]. Problem record: `problems/split-prime-pattern-scalar-only.md`.

1. Question and answer

Request (user, 2026-11-07): a novel ECDLP algorithm based on something unique and not tried before, for example the prime-splitting pattern observed in the published prime curves (uploaded report `split_prime_report.pdf`, sha256 `d38be158…7a0c8`, "Split-prime deficit in published NIST / X9.62 prime curves").

Answer. The pattern is real: recounted here from the curve parameters, every split count and z the recount can compare agrees with the report, while the report's calibrated bands and its $L(1,\chi)$ normalisation are not reproduced and one of them is questioned (§3). It cannot feed an ECDLP algorithm below the generic bound on these curves: for every prime $\ell \neq n$, the objects it governs (ℓ-torsion, Frobenius eigenlines, the two rational ℓ-isogenies of a split ℓ, the class group of the Frobenius order, the endomorphism ring) reach the prime-order group $E(\mathbb{F}_p)$ only as multiplication by a known scalar or as an isomorphism that preserves the discrete logarithm (§2). This agrees with the lab's isogeny-class invariance framing (`research/isogeny_class_difficulty_20261008`) and with the measured null on index-calculus yield across a class (`research/isogeny_class_ic_yield_20261102`). Inside the generic model the pattern's one ECDLP-adjacent consequence is a lower degree of the horizontal expander used by random self-reduction (§4); whether it correlates with coordinate-method yield between classes is untested (proposal P3, §5). Candidate directions that survive the obstruction, with their cheapest tests, are in §5; none is a claim.

Scope. The ten prime-order curves prime192v1 (P-192), prime192v2, prime192v3, secp224r1 (P-224), prime239v1/v2/v3, prime256v1 (P-256), secp384r1 (P-384), secp521r1 (P-521); p and n from `openssl ecparam -param_enc explicit` (OpenSSL 3.0.13), cofactor 1. Objects: $t = p + 1 - n$, $\Delta = t^2 - 4p$, $K = \mathbb{Q}\left(\sqrt{\Delta}\right)$, and for each odd prime ℓ the Kronecker symbol $\chi(\ell) = \left(\frac{\Delta}{\ell}\right)$: split (+1, Elkies), zero ($\ell \mid \Delta$: ramified in K or dividing the conductor of $\mathbb{Z}[\pi]$, not separated here, e.g. $3 \mid f$ for P-224; the tables keep the report's label "ramified"), inert (−1, Atkin).

Sources checked. Prior in-repo form of the obstruction, found after the first commit: `RESEARCH_ISOGENY_FIBERS_ECDLP.md` §15.1, Theorem 15.1 (isogeny-augmented generic group model: oracles for every rational isogeny of the class with their scalars leave a Q-query algorithm at success probability at most $(Q+2)^2/r + 1/r$; [pcc:unverified] there); §2 below is its specialisation and claims no novelty. The statistic itself was already computed by the lab for the published and omitted curves (`research/nist_skip_criterion_20261025/strict_elkies_20261029/results/at691.tsv`), with the same ten counts as §3. KB unavailable (the `research-kb` MCP server is not loaded; `scripts/research-kb` finds no executable; the lead atlas checkout is absent). Repository paths read: `research/isogeny_class_difficulty_20261008/README.md`, `research/isogeny_class_ic_yield_20261102/p2_prime_analogues/RESULTS.md` and `report/REPORT.md`, `problems/prime-scalar-exact-splitting.md`, `problems/tree-glued-pdp-separator-coupling.md`, `problems/curve25519-family-algebraic-ic-tower.md`, `docs/endomorphism-rules.md`. Literature from memory, primary sources not re-read this session: Lenstra 1996 ($E(\mathbb{F}_q) \cong \mathcal{O}/(\pi - 1)$ as $\mathcal{O}$-modules), Jao–Miller–Venkatesan 2005 (random self-reducibility across an isogeny class via expander walks over split ℓ, under GRH), Shoup 1997 ($\Omega(\sqrt{n})$ generic bound), Schoof–Elkies–Atkin (split ℓ speed up point counting only).

2. The obstruction (derived; no Lean proof yet; prior form: Theorem 15.1)

Setting. $E/\mathbb{F}_p$ ordinary, $\#E(\mathbb{F}_p) = n$ prime, cofactor 1, Frobenius π with $\pi^2 - t\pi + p = 0$, $\mathcal{O} = \mathrm{End}(E) \supseteq \mathbb{Z}[\pi]$, $N = (\pi - 1)$ the ideal of $\mathcal{O}$ of norm n. Standing assumptions made explicit after the red-team pass: E ordinary, so $\mathrm{End}(E)$ is defined over $\mathbb{F}_p$; $\gcd(n, \Delta) = 1$ (checked on all ten curves by the red team); $\mathcal{O}^\times = \pm 1$. The exclusion $\ell \neq n$ is a convenience: the two rational n-isogenies are $\pi - 1$ and $\pi - p$, the scalars 0 and $1 - p$.

Theorem (scalar-only action). Let ℓ be a prime different from n and p. Every $\mathbb{F}_p$-rational isogeny $\varphi : E \to E'$ of ℓ-power degree (each horizontal ℓ-isogeny of a split ℓ, their compositions around the class group, every vertical ℓ-isogeny) and every endomorphism $\alpha \in \mathcal{O}$ satisfy:

1. φ restricted to $E(\mathbb{F}_p)$ is a group isomorphism onto $E'(\mathbb{F}_p)$ and $\varphi([k]P) = [k]\varphi(P)$: a transported DLP instance has the same answer.
2. α acts on $E(\mathbb{F}_p)$ as multiplication by $s_\alpha = \alpha \bmod N \in \mathbb{Z}/n$, computable from the coordinates of α in $\mathcal{O}$ (for $\alpha = u + v\pi$, $s_\alpha = u + v$).
3. The Galois module $E[\ell]$, its Frobenius eigenvalues λ, μ, the fiber polynomials of φ, and every ℓ-pairing value involving a point of $E(\mathbb{F}_p)$ are independent of which point of $E(\mathbb{F}_p)$ is considered.

Proof sketch. (1) $\deg \varphi$ is prime to n, so $\ker \varphi \cap E(\mathbb{F}_p) = \{O\}$ and φ is injective on the finite group $E(\mathbb{F}_p)$; $E'(\mathbb{F}_p)$ has the same order, so φ is onto; it is a homomorphism, so it commutes with $[k]$. (2) $E(\mathbb{F}_p)$ is cyclic of prime order, so every endomorphism of it is a scalar; which scalar follows from Lenstra's isomorphism $E(\mathbb{F}_p) \cong \mathcal{O}/N$ under which $\pi \mapsto 1$. (3) $E[\ell]$ and $E(\mathbb{F}_p)$ are Galois modules of coprime order; a bilinear pairing $t(T, P)$ with $\operatorname{ord} T = \ell$, $\operatorname{ord} P = n$ satisfies $t^\ell = t^n = 1$, so $t = 1$; the fiber of φ above $\varphi(P)$ is $P + \ker \varphi$ with the Galois structure of $\ker \varphi$. ■

Corollary. An algorithm whose only operations are the group law, the maps above and point-independent Galois data is a generic-group algorithm with scalar oracles: every oracle answer encodes a known degree-one polynomial in the unknown log, so Shoup's simulation argument applies verbatim (the proof of Theorem 15.1) and gives $\Omega(\sqrt{n})$ oracle calls. So the split-prime pattern alone, for any cutoff and any deficit, cannot yield a sub-generic algorithm on a prime-order curve; by the JMV self-reduction every curve reachable along the horizontal edges that the split primes provide is exactly as hard.

Not covered. Anything that reads $\mathbb{F}_p$-coordinates non-linearly: summation polynomials, division polynomials, and (for small embedding degree only, which these curves lack) the n-pairing. These are the lab's index-calculus tracks. The pattern can enter them indirectly by choosing which curve of the class to decompose on; the lab measured that choice null on 2026-11-02 (P2: ICC not distinguishable from zero at 24–32 bits, planted-20 % power 0.99–1.00 in the informative cells, two 24-bit $m = 3$ cells not informative, one suspected curve effect not replicated). Two presentations sit outside the theorem's hypotheses, as Theorem 15.1′s reading says: the n-pairing (embedding degree above 20,000 on all ten curves, checked by the red team) and the p-adic lift: since $\pi \equiv 1 \bmod N$, every point of $E(\mathbb{F}_p)$ has a computable canonical lift with p-adic coordinates, a third presentation covered neither by the theorem nor by the measured null. For $n \neq p$ the p-adic elliptic logarithm vanishes on torsion (why Smart's attack needs $n = p$); that this closes the presentation is an assertion here [pcc:unverified], recorded in the problem record.

The reformulation that looks like a way in. By Shimura reciprocity the DLP k is the element of $\mathrm{Gal}(K_N/H) \cong (\mathcal{O}/N)^\times / \pm 1$ mapping the canonical-lift torsion point $\tilde{P}$ to $\tilde{Q}$, and the Artin symbol of a principal ideal (α) acts as $\alpha \bmod N$. Writing k as a product of known scalars s_α of smooth principal α is a multiplicative DLP in $\mathbb{F}_n^\times$, feasible by index calculus — but only on the residue k, which is the unknown. The group is presented by coordinates, never by residues in $\mathcal{O}/N$, so "index calculus in $\mathcal{O}$" has no first step; enumerating smooth α and comparing $\alpha(P)$ with Q is baby-step giant-step with an unusual giant-step set and the same $\sqrt{n}$ cost.

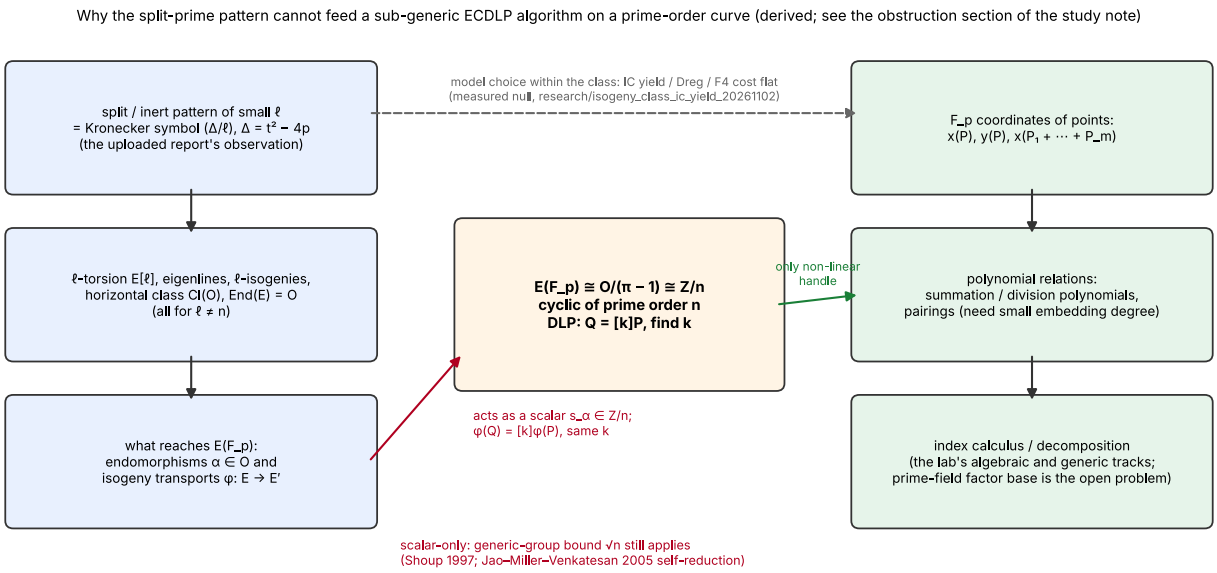


Figure 1: Where the pattern lives (left), the group it must reach (centre), and the only non-linear handle on that group (right). Derived; the dashed link is the route measured null on 2026-11-02.

3. Independent recount of the deficit (measured, native)

Method. `tools/splitcount.rs` (dependency-free Rust, `rustc -O`, 0.3 s). For each odd prime $\ell \le 200000$ it reduces the hex parameters p, n modulo ℓ, forms $t \equiv p + 1 - n$, $\Delta \equiv t^2 - 4p$, and takes the Legendre symbol by Euler's criterion. Baseline (the report's model, re-derived in the source comment): Frobenius mod ℓ uniform among matrices of $\mathrm{GL}_2(\mathbb{F}_\ell)$ with determinant p, so $P(\text{split}) = (\ell - 2 - \chi(p))/(2(\ell-1))$, $P(\text{ramified}) = (1+\chi(p))\ell/(\ell^2-1)$; the trace class $\tau \equiv p+1$ (which would give $\ell \mid n$) is removed and the rest renormalised. $z = (\text{observed} - \sum P)/\sqrt{\sum P(1-P)}$ over odd primes $\ell \le B$; twelve cutoffs as in the report. Output `data/recount.json` (sha256 `e84f89d3…7d98`; source sha256 `cb1dc02c…7550`). No $\ell \le 200000$ divides any n.

curve	split (obs.)	expected	sd	z @ 691	z @ 5,000	z @ 50,000	L-ratio, ℓ ≤ 200k
prime192v1 (P-192)	60	59.7	5.51	+0.05	+0.28	+0.80	0.93
prime192v2	58	59.7	5.51	−0.31	−2.20	+0.66	1.20
prime192v3	57	59.7	5.51	−0.49	−1.12	−1.51	0.80
secp224r1 (P-224)	50	60.1	5.53	−1.82	−2.54	−1.50	0.64
prime239v1	42	59.7	5.51	−3.21	−1.28	−0.12	0.56
prime239v2	57	59.7	5.51	−0.49	−0.89	−2.07	0.86
prime239v3	57	59.7	5.51	−0.49	+0.35	−0.51	0.93
prime256v1 (P-256)	57	60.2	5.54	−0.57	−0.30	−1.42	1.27
secp384r1 (P-384)	54	59.9	5.52	−1.06	−0.59	−1.21	0.62
secp521r1 (P-521)	52	60.3	5.54	−1.51	+0.54	+0.54	1.01

Counts at $B = 691$ (124 odd primes). Pooled z over the ten curves, $(\sum \text{obs} - \sum \text{exp})/\sqrt{\sum \text{var}}$, at B = 100, 200, 500, 691, 1000, 2000, 5000, 10000, 20000, 50000, 100000, 200000: −2.71, −1.81, −2.98, −3.13, −1.98, −2.16, −2.45, −2.12, −2.26, −2.00, −1.87, −1.79 (Stouffer $\sum z/\sqrt{10}$ agrees to two decimals). The last column is this study's normalisation of the partial Euler product $\prod (1 - \chi(\ell)/\ell)^{-1}$ over odd $\ell \le 200000$, as a ratio to $\exp E[\log \prod]$ under the conditioned model; it is not the report's "× typical" column: the two lowest (prime239v1, P-384) and the highest (P-256) agree, the rest of the order does not, and the values differ by about 0.78–0.80× on seven curves. Red-team finding F4 (plausible, `ideas/red-team.md`): the report normalises all ten curves to P-224-field bands, but for $p \equiv 2 \pmod 3$ fields (P-192, P-239, P-384) a prime-order curve has $\ell = 3$ forced inert, so the model's expected partial product is about 0.66 there against 0.84–0.86 for $p \equiv 1 \pmod 3$; the report's "prime239v1 × 0.44, P-384 × 0.49 outside the 95 % band" is at least partly a field-mismatch artefact (field-matched: 0.56, 0.62). An observation about the report, not a re-analysis.

Agreement with the uploaded report. Every split count at 691, every per-curve z at 691, 5,000 and 50,000, and every pooled z at the twelve cutoffs matches the report's table and list to the printed precision [pcc:unverified]. The report's random-curve calibration (4,267 SEA-counted curves) and its percentiles were not reproduced; no statement is made about them. Nothing here bears on the cause of the deficit; the report's own caveats (cutoff chosen after the fact, correlated cutoffs, min-of-three families, ten curves) stand.

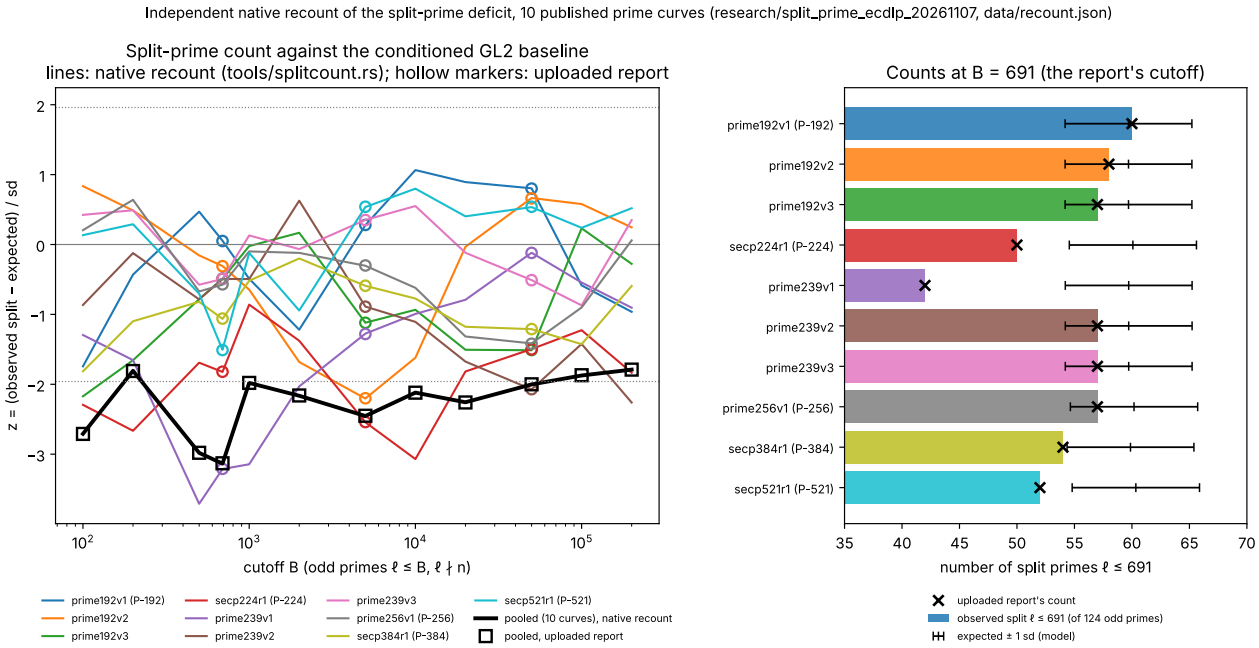


Figure 2: Left: z against cutoff for each curve (lines, native recount) with the report's values as hollow markers; black: pooled. Right: counts at $B = 691$ against the model expectation ± 1 sd. Source `figures/plot.py`, data `data/recount.json`; n = 124 primes at $B = 691$.

4. The one ECDLP-adjacent effect inside the generic model: horizontal expander degree

The JMV self-reduction walks the horizontal class along ℓ-isogenies for split $\ell \le B$; the graph has degree $2 \times \#\{\text{split } \ell \le B\}$. A deficit lowers the degree for a given B: at $B = 691$, P-224 has 50 split primes (degree 100) against 60 expected (degree 120), prime239v1 has 42 (degree 84). The walk needs a slightly larger B for the same mixing, i.e. a slightly larger transport cost; by §2 the destination curve is exactly as hard. The lab's protocol I-2 (`research/isogeny_class_difficulty_20261008/PROTOCOL-I2-transport-cost.md`) already prices transport; this study adds only the observation that the report's statistic is the degree of that graph. Outside the generic model this section claims nothing: Δ is a class invariant, so the within-class null cannot say whether the pattern correlates with coordinate-method yield between classes (red-team F1); that is proposal P3 in §5.

5. Candidate novel directions that survive the obstruction

From the idea-generator (Sonnet) and idea-synthesist (Opus) passes of this session (full outputs in `ideas/`). Every item is **proposed**, none is run, none is a claim. Both passes independently re-derived the §2 obstruction and neither found a mechanism in which the split-prime deficit enters: nine candidate uses were killed on paper (isogeny-pullback towers on a prime-order curve, where every isogeny fiber has exactly one $\mathbb{F}_p$-point; twist-torsion translations, defined over $\mathbb{F}_{p^2}$; class-group label cycles; smooth-norm endomorphisms as x-line maps; torus-coset bases with a trace map; Kummer-plus-Dickson unions; cross-target Gröbner; twist or Tate leakage; the CM lift and ray class field, whose isomorphism is defined only up to $\mathrm{Gal}(K_N/K)$, which acts as every scalar). The directions below are coordinate-side proposals for prime-order curves in the NIST field shape, with the proposers' own priors.

id	mechanism	cheapest native test	kill line	proposer's prior
P1 RR-Dickson torus	Separator-free Riemann–Roch norm-divisibility system with a Dickson-torus membership tower, prime-order curves	Rust emitter into msolve 0.6.5; RR-Dickson, T-Dickson, RR-Kummer, T-	$D_{\mathrm{RR(Dickson)}} \ge D_T(\mathrm{Dickson})$ everywhere; or RR	0.55 stage-level at $m = 3$; 0.05 repo criterion

	with $p \equiv 3 (\mathrm{mod}\, 4)$, $2^t \mid p+1$ (NIST shape, no Kummer tower). Prior: in-repo `round7_rr`, `pkm_dickson_torus_20261015`; PKM 2016, Nagao 2013.	Kummer on one planted instance at T24 and T31; $m = 3$, $t = 3$–5; about 1 h	Dickson vs Kummer differ by ≥ 2; or the RR-Kummer control fails to reproduce round7	($m \geq 5$, blocked by 15 GB); 0.01 end-to-end
P2 twist-pure coset	Select the 2-power coset whose 2^t abscissae are mostly on E; derived purity lemma (twist parts of an x-only relation sum to O) and decay law $c_{\mathrm{sel}}/c_{\mathrm{mean}} \approx 1 + \sqrt{2 \ln C / 2^t}$. Prior: `curve25519_frame_search_20261103`; FGHR 2014.	PKM oracle crate and ecbench: random vs argmax coset at equal (m, t), exact $m = 2$ coverage, restricted-base control; 2^{20}–2^{32}; hours	F4 cost differs > 10 % between cosets; coverage ratio off > 20 %; lemma fails on exhaustive S_3 enumeration at $p \leq 10^4$	0.85 toy F4 work per relation; 0.05 cold end-to-end S; 0 at cryptographic size
C1 presentation × arithmetic	Original model, dense random shift, curve-sparse shift ($a = -3$: $x = X \pm 1$ kills the linear term), tower-sparse shift; $j = 0/1728$ twists, 64 $a = -3$ curves, 64 with $-a/3$ square; $p = 786433$ and a 24-bit prime. Lemma: Semaev polynomials are weighted-homogeneous (x:1, a:2, b:3), top form curve-independent. Prior: `isogeny_class_ic_yield_20261102` P1.	`pkm_ss_tower f4`; 0.5–5 CPU-hours after 1–2 days of code	arithmetic hypothesis dies if a $j = 0/1728$ difference vanishes under the dense shift; lever dies if sparse/dense mul-add ratio > 0.90 on ≥ 2 of 4 cells or D rises	0.02 arithmetic effect; 0.03 engineering gain (toy)
C2 j-line census	Every prime-order curve at $p = 786433$, per trace (about 250 Frobenius fields), against split count, $L(1, \chi)$, h, conductor; reaches the CM strata random samples miss. Prior: `isogeny_class_difficulty_20261008`, `p2_lift_defect_20261101`.	Rust census; 1–25 CPU-hours	null rejected only if an unexplained exception reproduces on fresh targets and is enriched in one K stratum (Bonferroni $\alpha = 0.0025$)	0.01 K-related; ≤ 0.005 baseline gain
P3 seed negative control	Cross-trace covariate test: 2,000 prime-order curves per prime at 2^{24}, 2^{28}, 2^{32}; rho.negation S and ic.pipeline yield against E_{691}, partial $L(1, \chi)$, division-polynomial root counts, conditioned on n. Prior: `isogeny_class_ic_yield_20261102` P2, `nist_skip_criterion_20261025`; JMV 2005.	ecbench; planted +2 % yield positive control, shuffled-trace negative control; hours at 24–28 bits	a replicated partial effect ≥ 1 % with FDR < 0.01 at two primes would reject the null and make the seed a lead	0.02 non-null; 0 baseline gain

Reading. The only proposals with a non-trivial prior (P1, P2) transfer in-repo levers to the prime-order field shape and are not new mechanisms; their priors for the repository's own criterion are 0.05 or lower. The two that touch the seed (C2, P3) are falsifiers whose expected outcome is a null that closes it for the coordinate tracks too. Any run follows the lab's rules (frozen task card, `.resume/RESUME.md`, bench record, pricing at 0, 0.1 and 1 GAE, `controlled-experiment` controls). Nothing is scheduled.

6. Status of every statement

statement	status
Split counts, expectations, z at twelve cutoffs for ten curves (§3)	measured natively from public parameters; agrees with the uploaded report; not independently checked by a second producer [pcc:unverified]
Theorem and corollary (§2)	derived (proof sketch); no Lean proof; no identity certificate (not a bilinear or summation identity)
"Model choice within the class is null"	measured by the lab on 2026-11-02 at 24–32 bits; cited, not re-measured
Expander-degree effect (§4)	derived from the counts; transport cost not measured here
Candidate directions (§5)	proposed; not run
Red-team review of §2–§4 (`ideas/red-team.md`)	done on commit 5e9c756773; mechanics confirmed (tool rebuilt byte-identical, OpenSSL parameters, independent big-integer recount with zero mismatches, baseline brute-forced for $\ell \leq 13$, at691 matched); findings F1–F8

	folded in; F4 on the report's normalisation is the reviewer's plausible finding, not re-analysed

Files. `tools/splitcount.rs`, `data/recount.json`, `data/uploaded_report_values.json`, `figures/plot.py` with `recount_z.{svg,png}` and `obstruction_diagram.{svg,png}`, `report/REPORT.typ` (this document), `ideas/` (the idea and red-team passes' outputs), `problems/split-prime-pattern-scalar-only.md`. **Next steps.** A Lean statement of §2 (1)–(3); of the §5 candidates, the one with the best prior under a frozen protocol on 24–32-bit toy classes with the controls of `controlled-experiment`; the report's own pre-registered number-theory follow-ups only if the owner wants the fingerprint question answered.